

Tech revolutions need time

NOT TOO LONG AGO, I REMEMBER A TIME WHEN MOVIES HAD TO BE RENTED ON VHS TAPES FOR ₹10 PER DAY. You'd discover new music on radio shows, where saving your favourite song and building your own playlist meant trying to find a double cassette tape recorder to create your own mixtape. Where analogue ruled supreme, digital was merely a dream; when my hair gleamed, and my trouser seams didn't scream. My apologies, I digress...

Before long the internet came and sped up the whole process, not to mention made it a lot more convenient too – whether you wanted to update your music or movie library. Sure, you still had to jump through some hoops, but entertainment moved away from cassette tapes and circular discs to simple point and click actions on your PC and smartphone screens.

The internet single-handedly modernized our personal and professional lives – whether it's ordering food or groceries online, video conferencing during the past two years of remote work, or seeking medical help via online doctor consultations when traveling in-person isn't an option. These are just a few of the myriad ways we are increasingly getting used to doing everything at the click of a button – always trying to reduce the time spent in getting what we want, with virtually no blood, sweat or tears spent along the way.

As the internet gradually started reaching more and more people around the world, it naturally led to an increase in pace of online innovation. Beyond what the internet fundamentally enabled (PCs and phones), consumer technology dealing with individual needs also modernized itself fairly quickly, you just have to look more closely inside our increasingly digital homes. Our previously dumb home appliances have become smarter over the years – from speakers, air conditioners, air purifiers, refrigerators and microwave ovens to robot vacuum cleaners and surveillance systems. They're getting connected to the internet so we can check their status or assign them small tasks, irrespective of where we are. If a process or product is for personal use, inside your home, chances are they have evolved multiple times over the past two-three decades, thanks to continual improvements made in communication technology.

The personal and domestic comforts witnessed over the past quarter of a century have given youngsters and people who've grown up in the internet age a false sense of urgency related to the speed of technological innovation itself. Beyond what the internet enables, the pace of innovation hasn't been as fast as we'd like in areas where technology and its implementation have a far more public impact. If you think about transportation alone, aeroplanes haven't fundamentally changed much on how they operate, despite getting smarter and more fuel efficient. Long distance trains still run on fossil fuels or non-renewable sources of energy. The materials used inside most batteries haven't changed in close to half a century, whether it's AAA alkaline batteries in our TV remote or lead-acid batteries used inside most automobiles – including those inside electric vehicles.

As much as we need the EV adoption curve to increase in order to combat the menace of global warming and climate change caused by burning fossil fuels, the world can't produce more EVs faster because of age-old battery technology that hasn't advanced enough. Some context to highlight the stakes – the pre-COVID world burned around 100 million barrels of oil each day, of which 1.5 million barrels of oil consumption was reduced each day of last year thanks to rise in global EVs usage, according to an Axios report. But a core component of electric batteries, lithium, continues to be in short supply – in fact it's going to remain scarce for at least the next five years. These critical raw material shortages will prevent automakers and governments around the world to reach their EV and carbon offset goals, even as startups try to disrupt and find new materials for storing energy. Quantum computers may herald a whole new age of discoveries related to material sciences, but quantum supremacy is still nowhere within reach despite 20+ years of research and development. Only if it were that easy or less time consuming, eh? 



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